

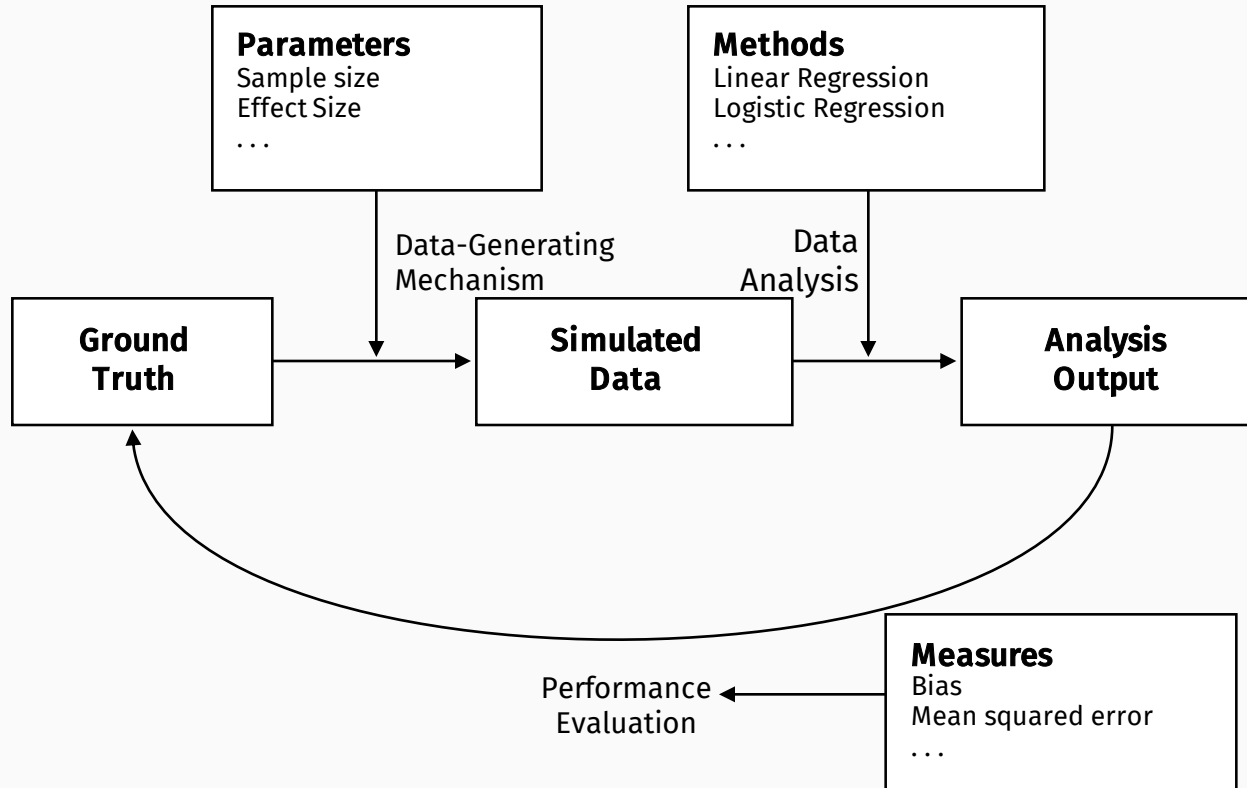
Towards Benchmarking of Psychological Time Series

FGME JuMi Retreat · October 2026

Björn Siepe

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Simulation studies





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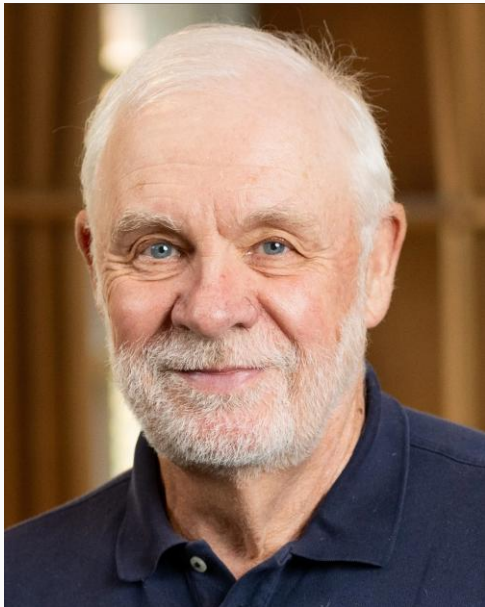
Psychological Methods

<https://doi.org/10.1037/met0000695>

Simulation Studies for Methodological Research in Psychology: A Standardized Template for Planning, Preregistration, and Reporting

Björn S. Siepe¹, František Bartoš², Tim P. Morris³, Anne-Laure Boulesteix^{4, 5},
Daniel W. Heck¹, and Samuel Pawel^{6, 7}

Issues in simulation studies



<https://src.isr.umich.edu/research/faculty-profiles/profiles/littleroderick-j/>

“ Good simulation studies are not given the respect they deserve. Often the **design is perfunctory and simplistic** [...] and results are over-generalized.

Roderick Little, 2013



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Psychological Methods

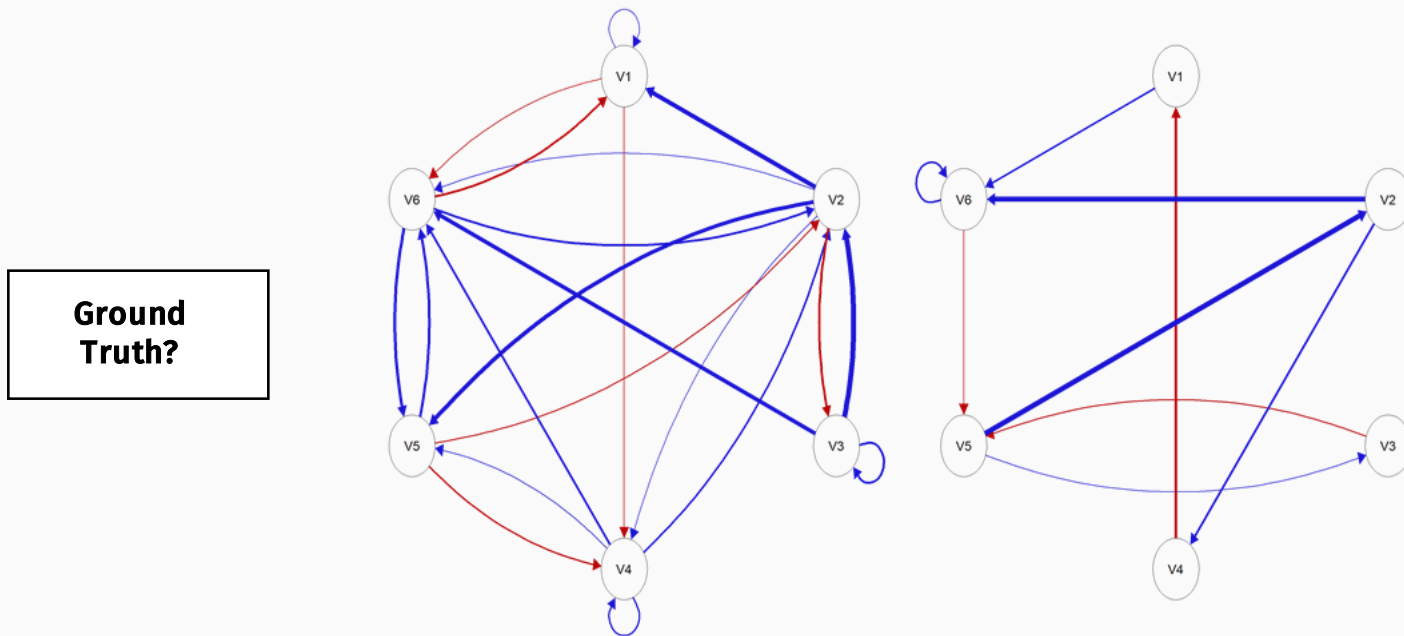
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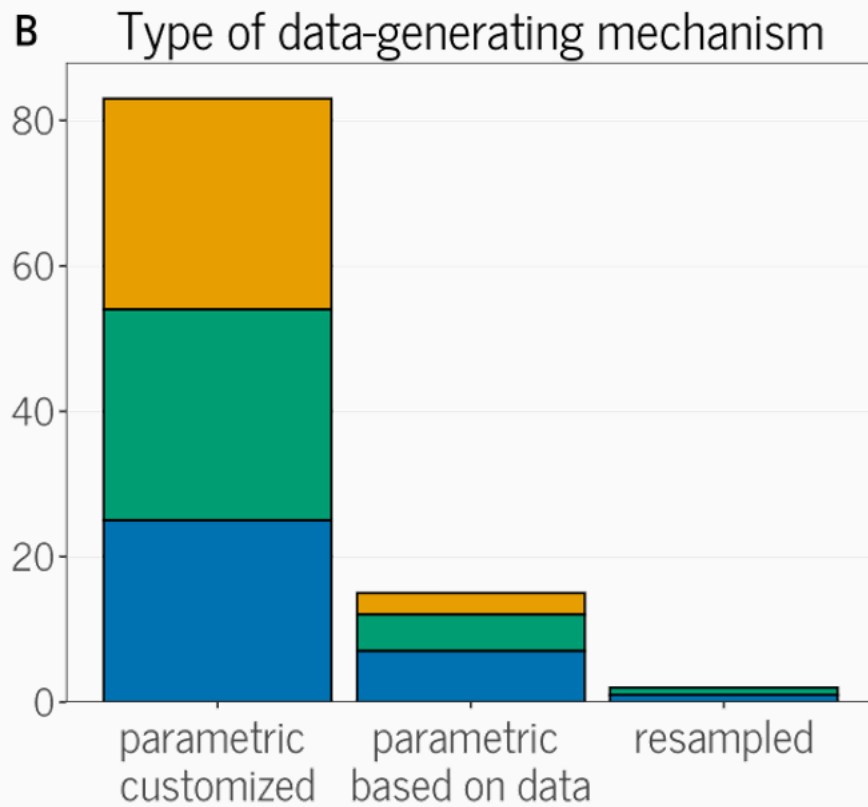
Few constraints on plausible DGMs

What is a useful/realistic data-generating mechanism?



→ Hard to understand when performance differences actually matter

What about *real* benchmarking?



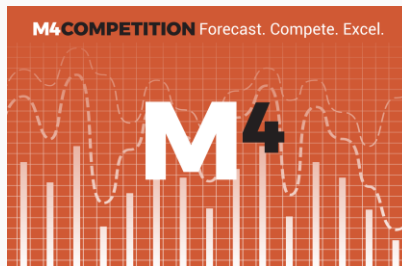
Psychology has been slow to adopt benchmarking

Computer Vision/ML



Source: Stenī Sebastian, Medium

Forecasting



Source: Rob Hyndman

Biostatistics

Robinson and Vitek *Genome Biology* (2019) 20:205
<https://doi.org/10.1186/s13059-019-1846-5>

EDITORIAL

Benchmarking comes of age

Mark D. Robinson^{1*} and Olga Vitek^{2*}

Why not in psychology?

(Rocca & Yarkoni, 2021)

1. Focus on in-sample performance

2. Lack of openly available data

Introducing openESM



Database

- Open-source
- Harmonized
- Rich metadata

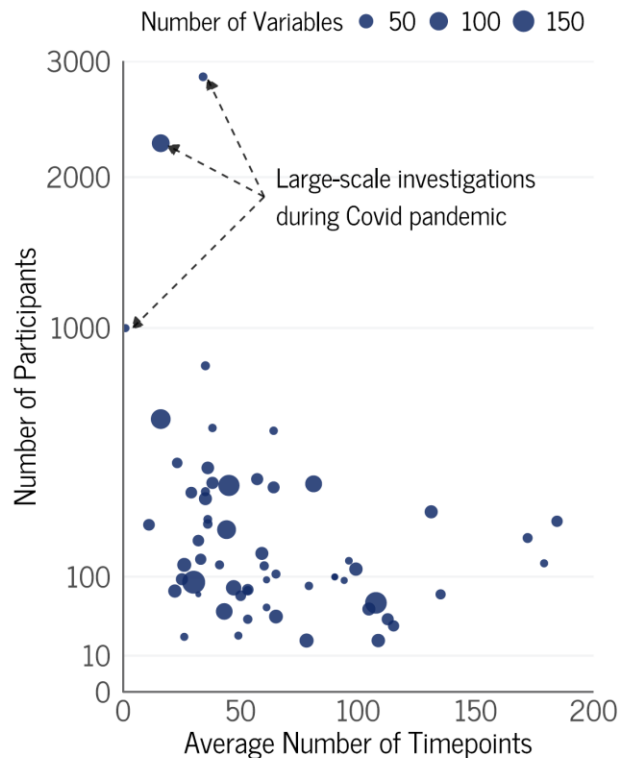
Scale

- **63** datasets
- **>18,000** participants
- **>1 million** observations

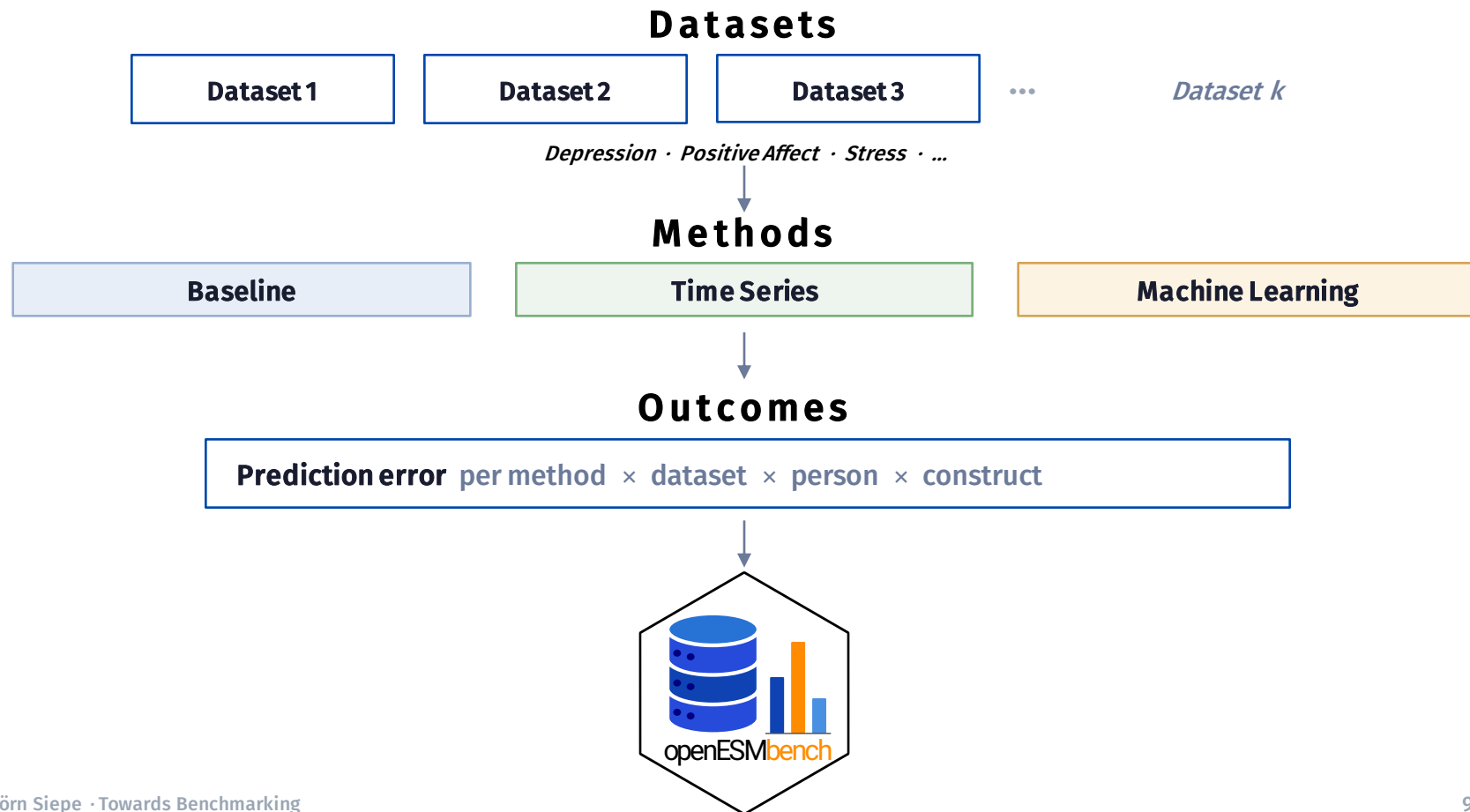
Access

- Website with search
-  and  packages

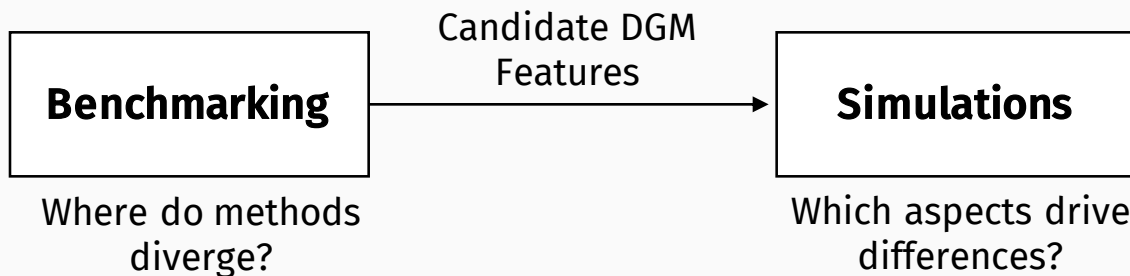
A. Timepoints, Variables, and Participants



Benchmarking Psychological Time Series Methods



Closing the loop: Benchmarking and simulations



RESEARCH ARTICLE | [Open Access](#) | [CC](#) [i](#)

On the role of benchmarking data sets and simulations in method comparison studies

Sarah Friedrich [✉](#) Tim Friede

First published: 21 February 2023 | <https://doi.org/10.1002/blmj.202200212> | [VIEW METRICS](#)



ADVANCED REVIEW | [Open Access](#) | [CC](#) [i](#)

Statistical Parametric Simulation Studies Based on Real Data

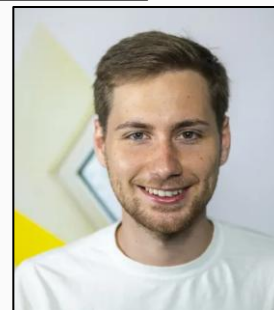
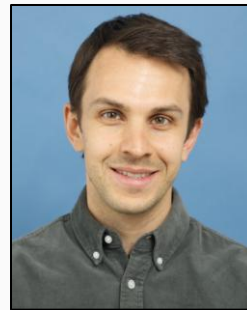
Christina Sauer, F. Julian D. Lange, Maria Thurow, Ina Dormuth, Anne-Laure Boulesteix [✉](#)

First published: 19 August 2026 | <https://doi.org/10.1002/wics.70074> | [VIEW METRICS](#)

Takeaways

1. Psychology is **ready for benchmarking**
2. Let's **not limit ourselves** to one way of investigating methods

Co-Authors



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Thank You!



<https://openesmdata.org>



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